

FREE TOOLS FOR DATA PROFESSIONALS

THE DATA PROFESSIONAL'S POWER TOOLKIT

Four free tools to capture, visualize, execute, and optimize your workflow.

TODAY'S TOOLCHAIN

Capture

Obsidian 

Visualize

Excalidraw 

Execute

VS Code 

Optimize

PowerToy 

No vendor pitches.
Just practical workflow wins.

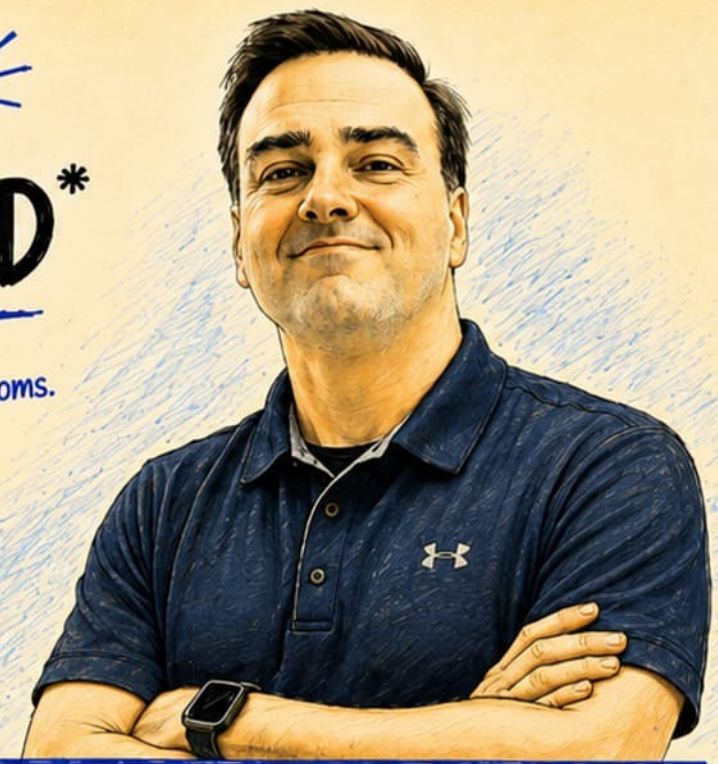
HI, I'M JASON

THE DAX SHEPHERD*

*More semantic models, fewer sitcoms.



Honored to be recognized by Microsoft.



I HERD THE DATA

-  Microsoft MVP
-  Nashville, Tennessee
-  Fabric + Power BI
-  Teaching + Speaking



MY FAVORITE MODEL

TARDIS

TOOLS & ALLIES



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Day of Data

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Bronze Sponsor



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All slides, resources, and demo files
available:

THE DAX SHEPHERD

thedaxshepherd.com/presentations



The DAX Shepherd

thedaxshepherd.com/presentations

Standing on the Shoulders of Giants

Grateful for the people who share their knowledge, encourage us, answer questions, and help the rest of us learn faster.



Photo credit: Unknown Author, CC BY-ND

SESSION ROADMAP

1. Intro

Why these four, and why together

1

Intro

2

Obsidian

3

Excalidraw

4

VS Code

5

PowerToys

6

Wrap-Up

NOW

1. Intro

Why these four, and why together

Next: Obsidian

SESSION ROADMAP

2. Obsidian

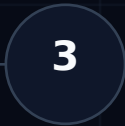
Capture notes, links, and ideas



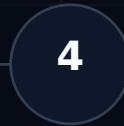
Intro



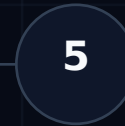
Obsidian



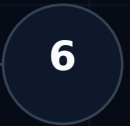
Excalidraw



VS Code



PowerToys



Wrap-Up

NOW

2. Obsidian

Capture notes, links, and ideas

Next: Excalidraw

SESSION ROADMAP

3. Excalidraw

Visualize systems and workflows



NOW 3. Excalidraw
Visualize systems and workflows

Next: VS Code

SESSION ROADMAP

4. VS Code

One editor for code, data, and diagrams



NOW 4. VS Code

One editor for code, data, and diagrams

Next: PowerToys

SESSION ROADMAP

5. PowerToys

Optimize everyday Windows workflows



NOW 5. PowerToys

Optimize everyday Windows workflows

Next: Wrap-Up

SESSION ROADMAP

6. Wrap-Up

Bring the toolkit together



NOW 6. Wrap-Up

Bring the toolkit together

SESSION ROADMAP

1. Intro

Why these four, and why together

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Wrap-Up

NOW

1. Intro

Why these four, and why together

Next: Obsidian

All Four Tools Are Free

Obsidian

FREE

Optional: Sync \$4/mo, Publish \$8/mo

Excalidraw

FREE

Optional: Excalidraw+ \$7/mo for cloud & collab

VS Code

FREE

No paid tier — fully free, forever.

PowerToys

FREE

No paid tier — fully free, forever.

Bring Your Own Installer

Tool	winget	choco	brew
Obsidian	Obsidian.Obsidian	obsidian	obsidian
Excalidraw	web app — no install	web app	web app
VS Code	Microsoft.VisualStudioCode	vscode	visual-studio-code
PowerToys	Microsoft.PowerToys	powertoys	<i>Windows only</i>

Already have a package manager? You're one command away.

Yes, We Need to Talk About AI



Tool	AI Superpower	How
Obsidian	Copilot plugin	Copilot plugin + local LLM support
Excalidraw	AI diagram generation	Generate diagrams from text descriptions
VS Code	GitHub Copilot	Copilot + Copilot Chat built right in
PowerToys	Advanced Paste	AI-powered clipboard transforms – local or cloud model

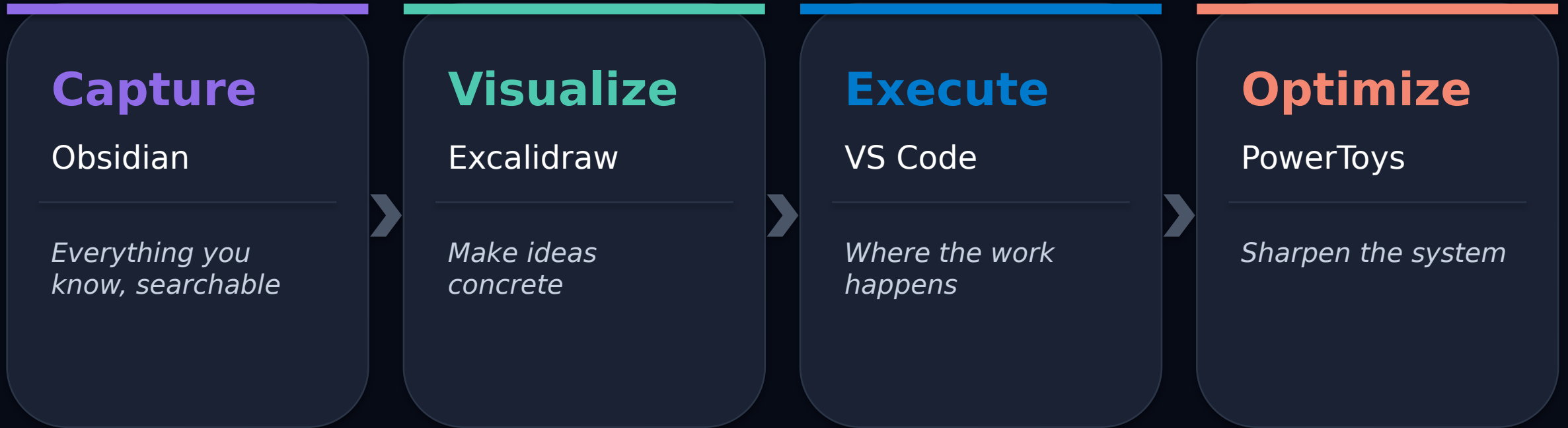
The tools were great before AI. They're better with it.

Every Tool is Extensible

Tool	Extension Model	Example
Obsidian	Plugins	Dataview, Calendar, Excalidraw
Excalidraw	Libraries	Pre-built data & architecture icon sets
VS Code	Extensions	Rainbow CSV, Jupyter, Python, Excalidraw
PowerToys	Modules	Enable only what you need

Start with the defaults. Extend to fit your workflow.

The workflow



Four tools, one loop — capture ideas, sketch them, build them, refine the workflow.

One keystroke to rule them all

Obsidian

Ctrl / Cmd + P

Open command palette

Excalidraw

Ctrl / Cmd + /

Open command palette

VS Code

Ctrl / Cmd + Shift + P

Open command palette

PowerToys Run

Alt + Space

Search anything, anywhere

One muscle — one keystroke, search anything, jump anywhere.

SESSION ROADMAP

2. Obsidian

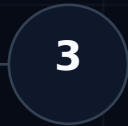
Capture notes, links, and ideas



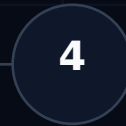
Intro



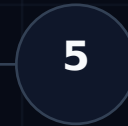
Obsidian



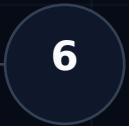
Excalidraw



VS Code



PowerToys



Wrap-Up

NOW

2. Obsidian

Capture notes, links, and ideas

Next: Excalidraw



01 · CAPTURE

OBSIDIAN

Capture notes, links, and ideas in one flexible knowledge base.

<https://obsidian.md/>

WHAT YOU'LL SEE

Fallout-themed vault · Dataview · Canvas · Templates & plugins

Don't pick a tool — pick a habit

- Beware the trap: spending more time **choosing** than **using**
- Your notes should serve *your work*, not become the work

If you have a system that works — **keep using it.**

Why Obsidian?

Your second brain for technical knowledge

Plain-text files you actually own — no vendor lock-in, open with any editor forever

Bidirectional links build a knowledge graph — see how ideas connect across your whole vault

Dataview turns your notes into a queryable database — filter and aggregate frontmatter like a spreadsheet

Conference Tracker

Submitted & Pending

Conference (3)	Closed	Expected Notification	Event Date	Location
Day of Data Orlando 2026	August 10, 2026	August 24, 2026	October 17, 2026	Orlando (Sanford), FL
Day of Data PNW 2026	July 31, 2026	-	November 07, 2026	Vancouver, WA
Day of Data Seattle 2026	August 17, 2026	-	November 12, 2026	Seattle, WA

✓ Accepted - Upcoming Talks

Conference (3)	Event Date	Location	Session Type	Accepted Session(s)
Nashville Microsoft Community Day 2026	September 11, 2026	Nashville, Tennessee		Data on Fire: A Hands-On Intro to Spark in Fabric
FabCon Europe 2026 - Call for Speakers	September 28, 2026	Barcelona, Spain	-	Not Every Notebook Needs Spark: SQL-First Analytics with Pure Python
Community Summit North America 2026	October 11, 2026	Nashville, TN	Breakout (60 min)	Dataverse for Fabric Practitioner

A live Dataview query against my conference YAML frontmatter — no spreadsheet required.

Obsidian — Demo

Let's walk the vault.

SESSION ROADMAP

3. Excalidraw

Visualize systems and workflows



NOW 3. Excalidraw
Visualize systems and workflows

Next: VS Code



02 · VISUALIZE

EXCALIDRAW

Visualize systems and workflows quickly without presentation friction.

<https://excalidraw.com/>

WHAT YOU'LL SEE

AWS architecture · Microsoft Fabric diagram · a fun Fallout sketch

Why Excalidraw?

Diagrams at the speed of thought

- no UI friction, just sketch and ship

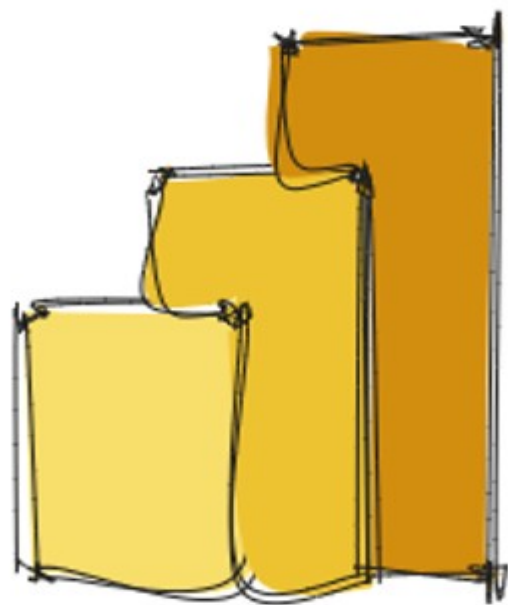
One-click icon libraries for any stack

- AWS, Azure, flowcharts, and more

Embeds directly inside Obsidian

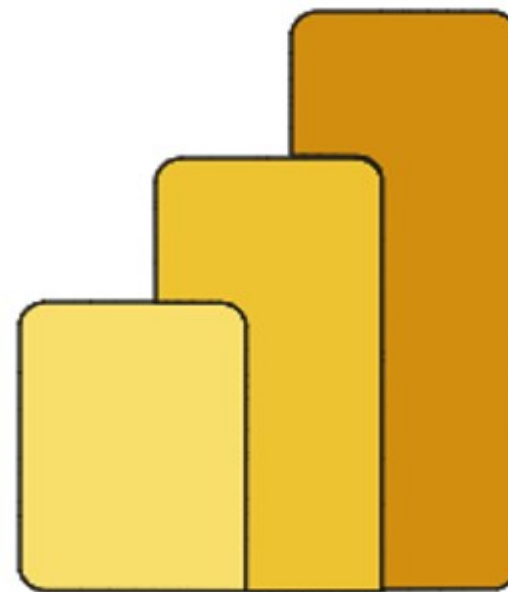
- diagrams live right beside your notes

Hand Drawn



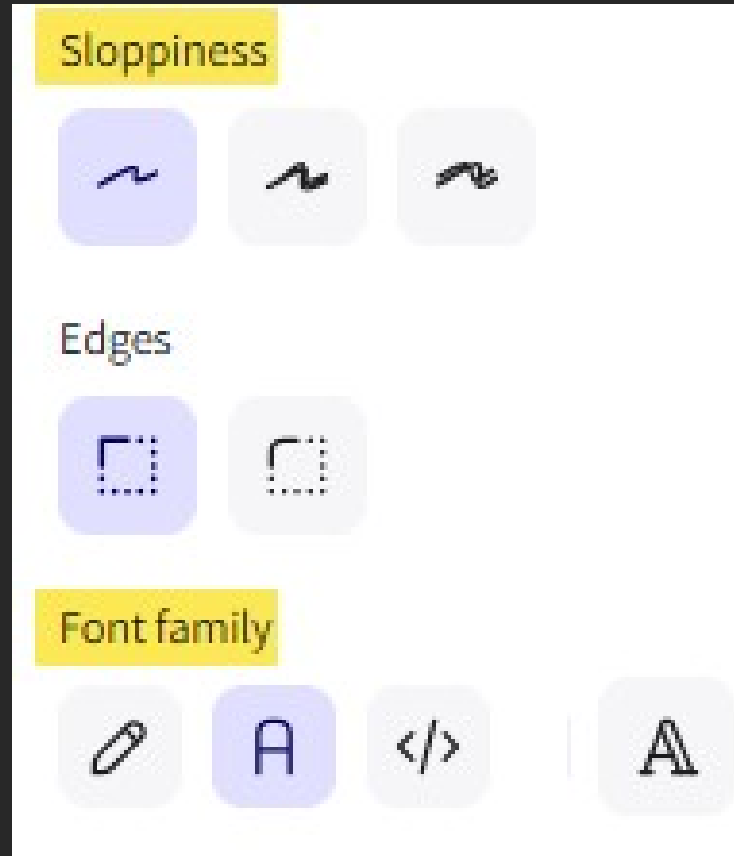
Power BI

Excalidraw



Original

Shape Properties



Quirk: Can't set sloppiness as a default

Workaround: Draw shapes, then **Ctrl+A** → adjust all at once

Excalidraw — Demo

Let's sketch something out.

SESSION ROADMAP

4. VS Code

One editor for code, data, and diagrams



NOW 4. VS Code

One editor for code, data, and diagrams

Next: PowerToys



04 · VS CODE

VS CODE (Visual Studio Code)

The editor many data pros already use
— for code, data, and diagrams.

<https://code.visualstudio.com/>

WHAT YOU'LL SEE

SQL on a Docker database · Jupyter notebook · Excalidraw diagrams

Why VS Code?

Free, cross-platform, 50,000+ extensions

- runs on Windows, Mac, and Linux

Built for data work out of the box

- Jupyter notebooks, Rainbow CSV, Python, and SQL all-in-one

GitHub Copilot as your AI pair programmer

- context-aware completions for code and text

VS Code — Demo

Let's pop open the editor.

SESSION ROADMAP

5. PowerToys

Optimize everyday Windows workflows



NOW 5. PowerToys

Optimize everyday Windows workflows

Next: Wrap-Up



05 · POWERTOYS

POWERTOYS

Optimize everyday Windows workflows with fast keyboard-driven utilities.

learn.microsoft.com/windows/powertoys/

WHAT YOU'LL SEE

FancyZones · Command Palette · Color Picker · Advanced Paste · ZoomIt

Why PowerToys?

Free & official — Microsoft-maintained, open source on GitHub

Modular — turn on only what you need; each utility runs independently

Keyboard-first — every utility has a hotkey, fits muscle memory

Power-user features Windows doesn't ship — window tiling, clipboard transforms, bulk rename, color picker

Constantly evolving — new utilities added regularly; community-driven

Note: some toys have been renamed or absorbed into Windows — e.g., Text Extractor now lives in Snipping Tool.

Try One Thing

You don't need to use all 20+ utilities.

Find the one that solves your most annoying daily problem.

Open PowerToys → pick it from the left menu → enable it.

For many of us it started with ZoomIt

Live Demo - What could go wrong?



Request failed: 403 Forbidden

Operation failed while accessing
_delta_log/_last_checkpoint

Request ID: 7f329b42-87ac-4cf2-9e91-27d94c87a680

Time: 2024-05-21 14:23:45 UTC

All PowerToys utilities



Launchers & quick actions

Advanced Paste

Paste clipboard into other formats — AI transfo **DEMO** available.

Command Not Found

Suggest WinGet packages for failed commands.

Command Palette

Launch commands, apps, and tools quickly. **DEMO**

PowerToys Run

Fast launcher for apps, files, and plugins. **DEPRECATED**

Quick Accent

Type accented characters without changing layout.

Screen Ruler

Measure pixels on screen with edge detection.

Text Extractor

OCR text from anywhere on screen. **DEPRECATED**

Windows & layout

Always On Top

Pin a window above the rest of the desktop.

Crop And Lock

Create a live cropped or thumbnail window.

FancyZones

Snap windows into custom layouts fast. **DEMO**

Peek

Preview file content without opening apps.

Shortcut Guide

Show available Windows shortcuts in context.

Workspaces

Launch saved app sets into arranged positions.

Light Switch

Auto-switch between light and dark themes.

All PowerToys utilities



Files & content

Color Picker

Grab color values from any pixel.

[DEMO](#)

File Explorer add-ons

Preview more file types inside Explorer.

File Locksmith

See which process is locking a file.

Hosts File Editor

Edit hosts entries with a safer UI.

Image Resizer

Resize images from the Explorer menu.

New+

Create files and folders from templates.

PowerRename

Bulk rename with search, replace, and regex.

System & input

Awake

Keep the PC awake without changing settings.

Environment Variables

Manage environment variable profiles.

Keyboard Manager

Remap keys and create custom shortcuts.

Mouse utilities

Find, highlight, or crosshair your cursor.

Mouse Without Borders

Use one keyboard and mouse across PCs.

Registry Preview

Preview and edit .reg files visually.

ZoomIt

Zoom, annotate, and record during demos.

[DEMO](#)

Newest additions



New in v0.100

Shortcut Guide

Rebuilt from scratch — a side pane that detects the active app and shows its shortcuts.

Command Palette: Extension Gallery

Browse, install, update, and remove extensions without leaving Command Palette.

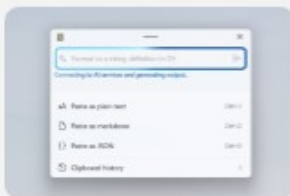
Command Palette: multi-monitor Dock

Each display gets its own Dock config, plus pinnable CPU, GPU, and battery metrics.

PowerToys ships new utilities every month — v0.100 landed June 10, 2026; current release is v0.100.2.

Options

Advanced Paste



Formats clipboard content into plain text, Markdown, JSON, and more. Advanced formatting can use an online or local language model endpoint.

[Learn more about Advanced Paste](#)



Advanced Paste

On ☒

Paste with AI



Paste with AI

Transform your clipboard content with the power of AI. A cloud or local endpoint is required.

[Learn more](#)

Off ☐ ▼

Activation & behavior



Open Advanced Paste window



Shift

V



Why these picks — daily drivers

Command Palette — universal entry point. One hotkey replaces hunting through Start menus and folders.

FancyZones — fills the biggest gap in Windows itself. Snap Layouts can't touch multi-monitor or custom grids.

Advanced Paste — shows the modern PowerToys: AI-assisted, format-aware, replaces ad-hoc scripts.

Why these picks — specialist tools

Color Picker — saves designers, devs, and slide-builders from screenshot-then-eyedropper workflows.

ZoomIt — purpose-built for presenting and teaching. Zoom, annotate, and record without third-party software.

PowerToys — Demo

Let's supercharge Windows.

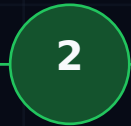
SESSION ROADMAP

6. Wrap-Up

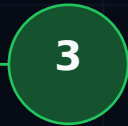
Bring the toolkit together



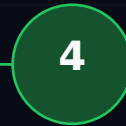
Intro



Obsidian



Excalidraw



VS Code



PowerToys



Wrap-Up

NOW

6. Wrap-Up

Bring the toolkit together

Pick one. Start today.

- Pick one tool today.
- Build a habit this week.
- Share it with your team.

Go Deeper

Tool	Documentation	Learn More
Obsidian	help.obsidian.md	<i>"Linking Your Thinking"</i> — Nick Milo on YouTube <i>How to Take Smart Notes in Obsidian</i> — Josh Duffney
Excalidraw	docs.excalidraw.com	Zsolt Viczián's YouTube channel Excalidraw+ blog (plus.excalidraw.com/blog)
VS Code	code.visualstudio.com/docs	Official VS Code YouTube channel VS Code Tips & Tricks (in official docs)
PowerToys	learn.microsoft.com/windows/powertoys	github.com/microsoft/PowerToys Scott Hanselman's PowerToys videos on YouTube

ool has a community worth joining — start with the official docs.

THANK YOU



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www.thedaxshepherd.com

Slides



Session Evaluation

